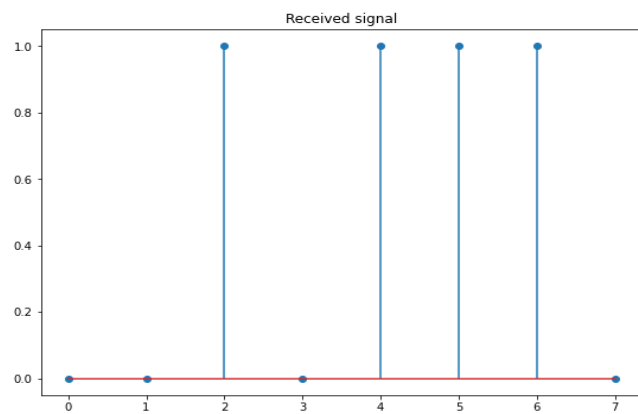
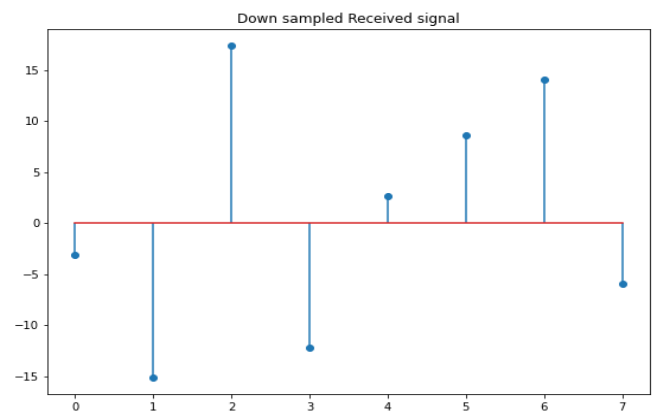
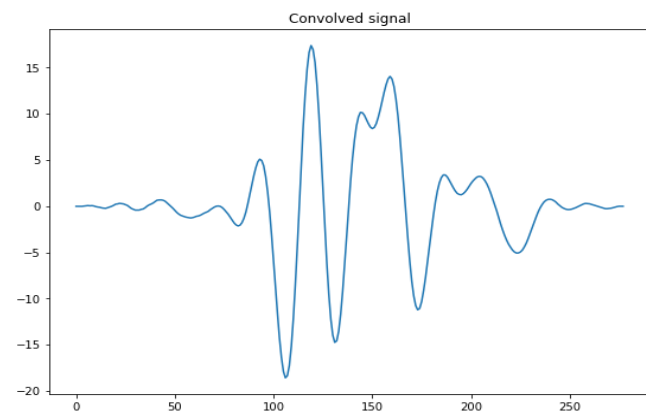
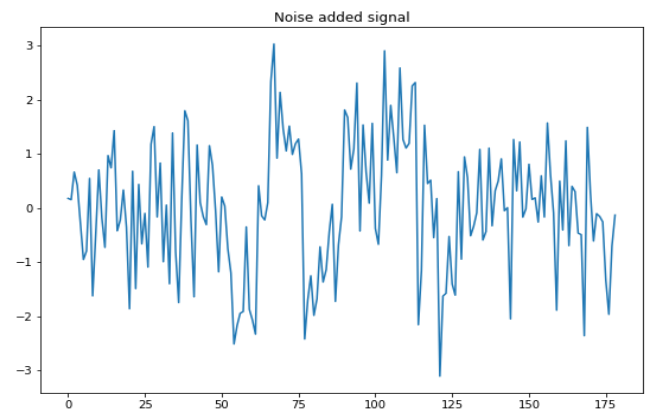
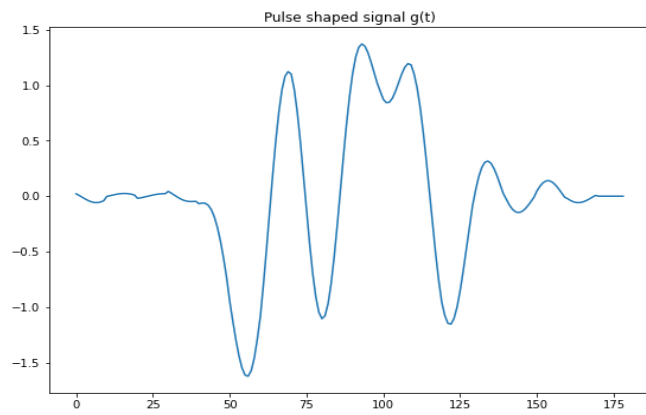
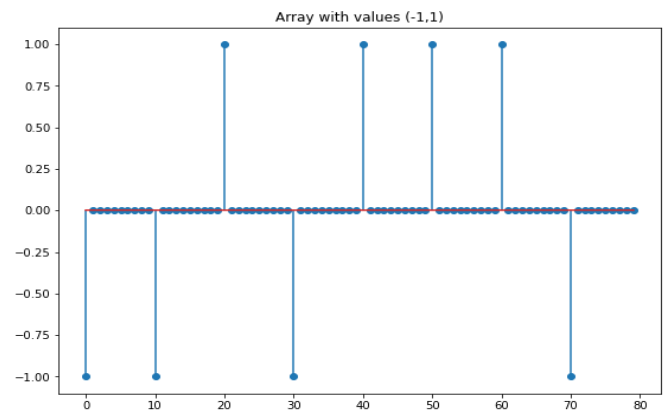
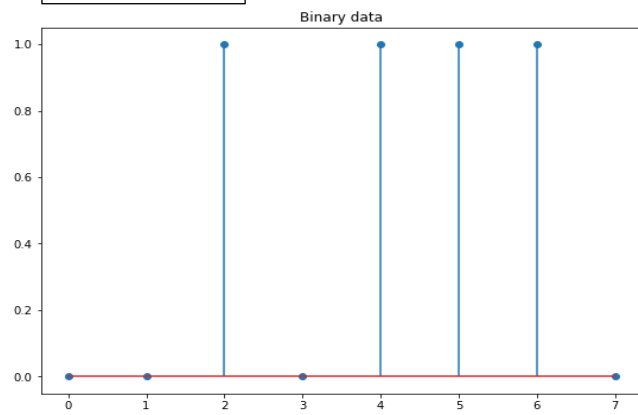


N=8



**PYTHON CODE**

```
import numpy as np

from commpy.filters import rrcosfilter as rrc

from numpy.random import standard_normal

import matplotlib.pyplot as plt

N,L=8,10

plt.figure(figsize=(20,30))

bits=np.random.randint(2,size=N)

plt.subplot(4,2,1)

plt.title("Binary data")

plt.stem(bits)

x=np.array([])

for i in bits:

    pulse=np.zeros(L)

    pulse[0]=i*2-1

    x=np.concatenate((x,pulse))

plt.subplot(4,2,2)

plt.title("Array with values (-1,1)")

plt.stem(x)

num,alpha,Ts,Fs=100,0.1,1,L

t,h=rrc(num,alpha,Ts,Fs)

y=np.convolve(x,h)

t1=np.arange(0,len(y))

plt.subplot(4,2,3)

#plt.figure(3)

plt.title("Pulse shaped signal g(t)")

plt.plot(t1,y)

snr=100

signal_power = np.sum(abs(y)**2) / len(y)

noise_power = signal_power / (10**(snr / 10))
```

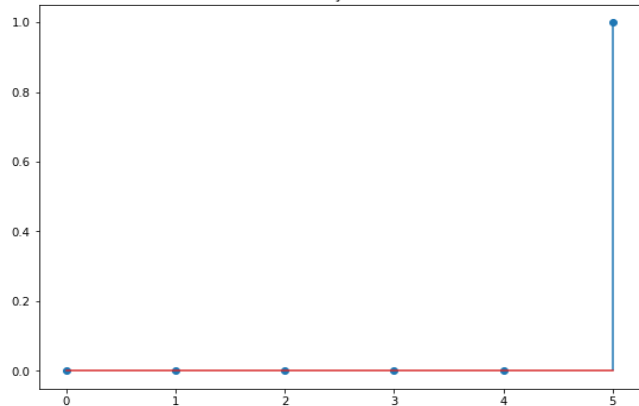
```

noise = np.sqrt(noise_power / 2) * standard_normal(y.shape) # Noise shape adjusted
r = y + noise
plt.subplot(4,2,4)
#plt.figure(4)
plt.plot(r)
plt.title("Noise added signal")
r1=np.convolve(r,h)
tr=np.arange(0,len(r1))
plt.subplot(4,2,5)
plt.plot(tr,r1)
plt.title("Convolved signal")
filter_delay=int((len(h)-1)/2)
r2=r1[2*filter_delay+1:tr[-1]-(2*filter_delay):L]
plt.subplot(4,2,6)
plt.stem(r2)
plt.title("Down sampled Received signal")
x_hat=(r2>0)
print(x_hat)
plt.subplot(4,2,7)
plt.stem(x_hat)
plt.title("Received signal")
BER1=np.sum(bits-x_hat)
BER=BER1/N
print(f"bit error : {BER1}")

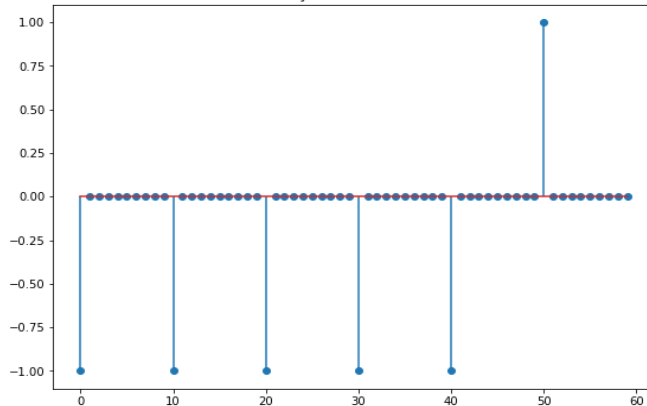
```

N=6

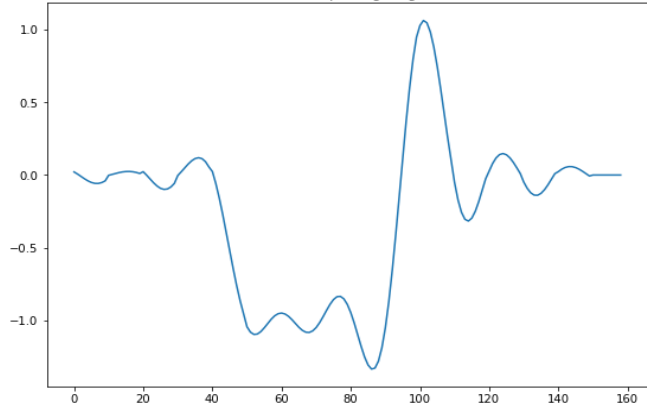
Binary data



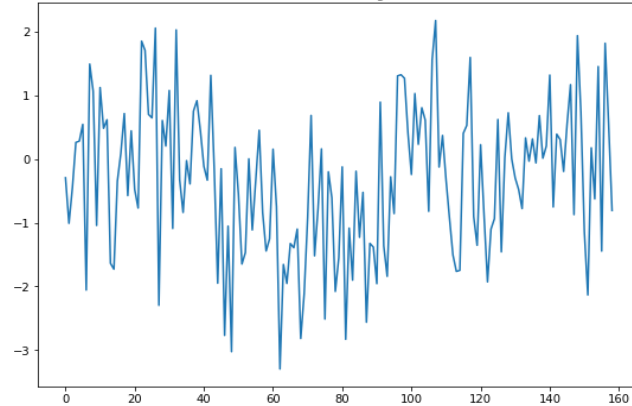
Array with values (-1,1)



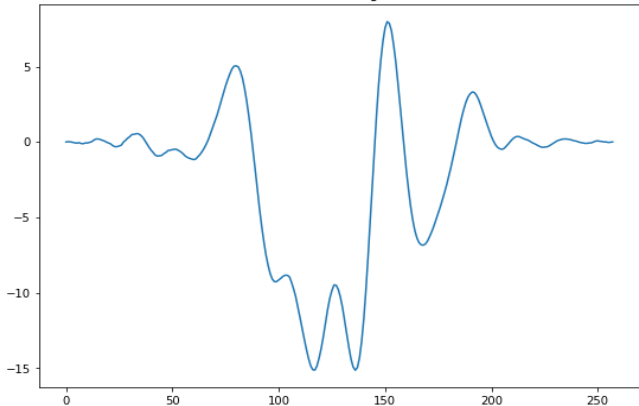
Pulse shaped signal g(t)



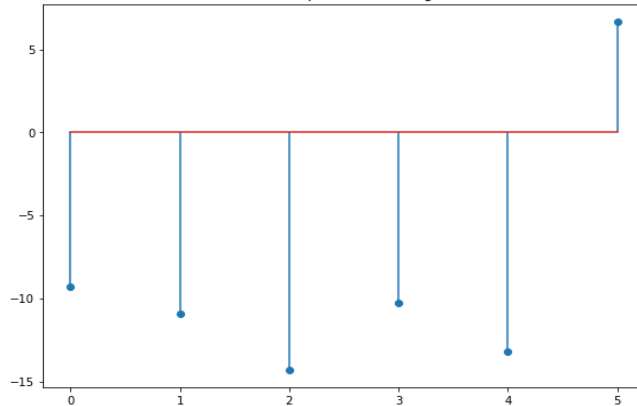
Noise added signal



Convolved signal



Down sampled Received signal



Received signal

